



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Subject : Database(SECD2523)

Section : 03

Task : Phase 2(P2) - Database Conceptual Design (ERD)

Lecturer : Dr. Haslina binti Hashim

Name	Matrics No
CHIA THUNG THUNG	A24CS0060
NG SHE LING	A24CS0285
TAN JIA YIE	A24CS0302
WONG YA JING	A24CS0211

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1.0 Introduction

This phase focuses on the conceptual database design of the Cengal Parcel Point(CPP), which was built to enhance the efficiency and data management at University Technology Malaysia (UTM). As online shopping continues to grow, CPP is required to handle a huge amount of parcels daily. The existing manual record system became inefficient, resulting in issues such as data inaccuracy, misplaced parcels, delayed notifications, and difficulty in tracking parcel status.

The proposed system introduces a centralized database that supports the automation and digitalization of CPP's operations. This database is designed to store, manage, and relate crucial information such as parcel details, sender and receiver information, locker assignments, staff records, and transaction history. Through an optimized Entity Relationship Diagram (ERD), the system ensures data consistency, integrity, and accessibility across all processes.

This conceptual design serves as the foundation for upcoming implementation of the system. It provides a clearer and centralized data structure used to support features like automation notifications, secure parcel collection, and real time status update. By using database principles, this phase helps to establish a reliable, efficient, and scalable data model that aligns with CPP operational goals, providing convenience and flexibility for both customers and staff.

This solution not only enhances service quality for students and staff but also reduces manual workload for CPP staff. With a focus on automation, security, and user experience, the Parcel Service System will deliver a scalable and modern platform capable of meeting CPP's current and future operational demands.

2.0 Data Flow Diagram (DFD)

2.1 Logical Data Flow Diagram (To-Be)

https://app.diagrams.net/#G1wDeJvFjk4KkwF4uHkbNpxh8rcrb7zqFf#%7B%22pageId%22%3A%227zJZ5BEME7tkb4WCI_1%22%7D

Context Diagram :

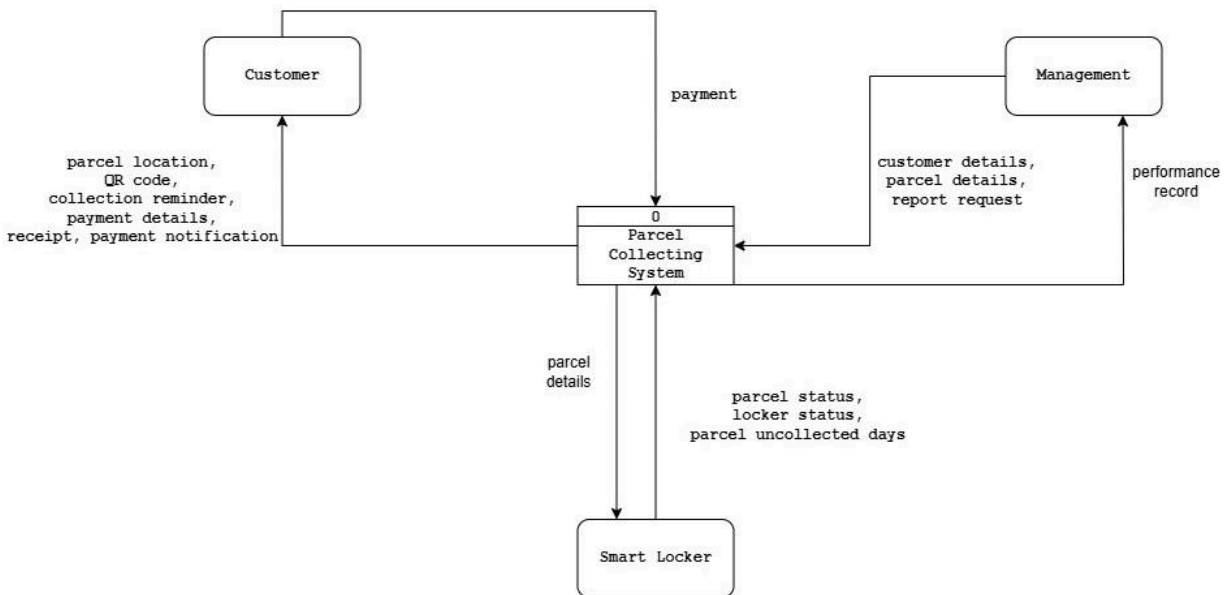


Figure 2.1.1: Context Diagram for TO-BE System

Diagram 0 :

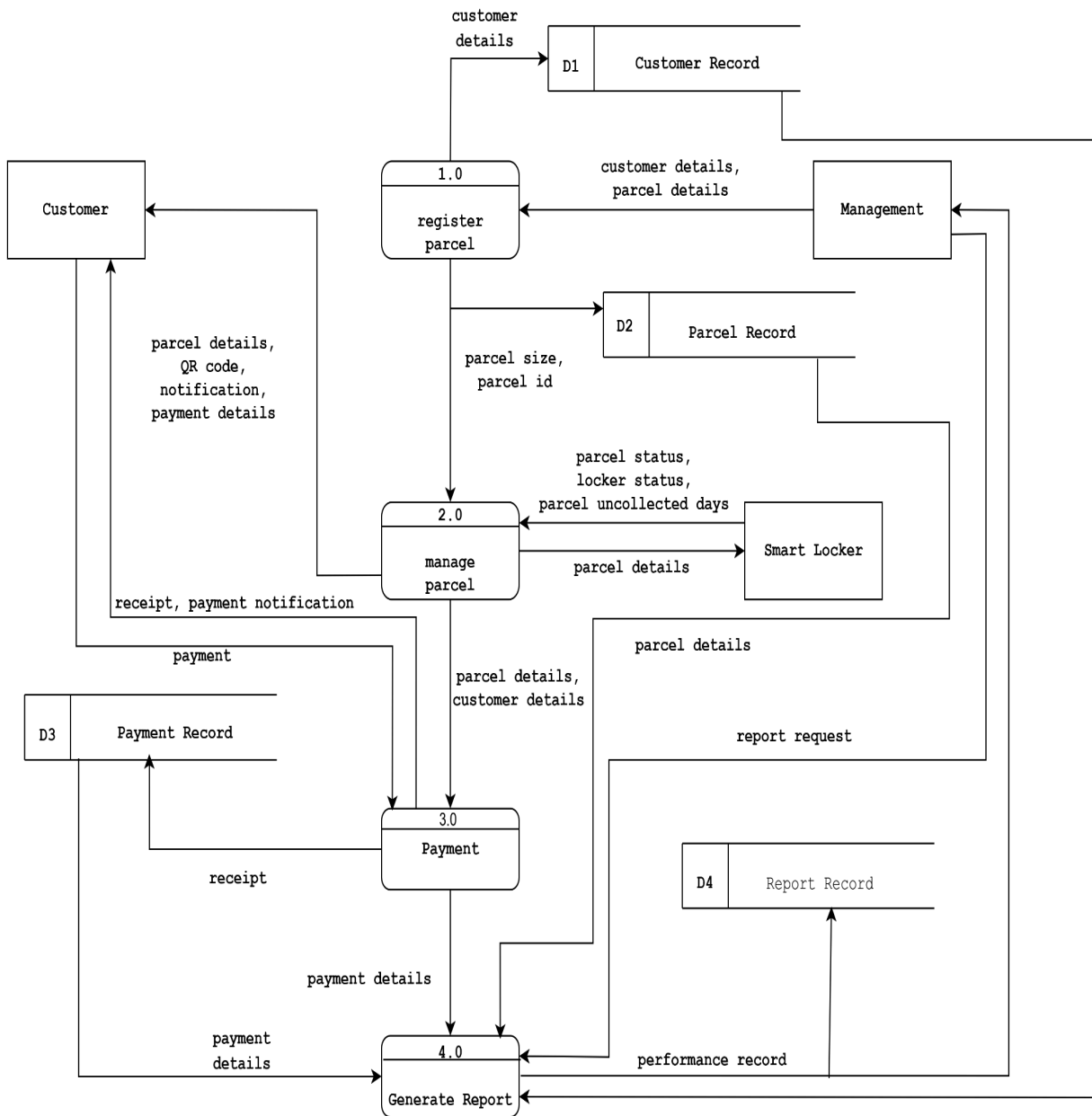


Figure 2.1.2: Diagram 0 for TO-BE System

Child Diagram :

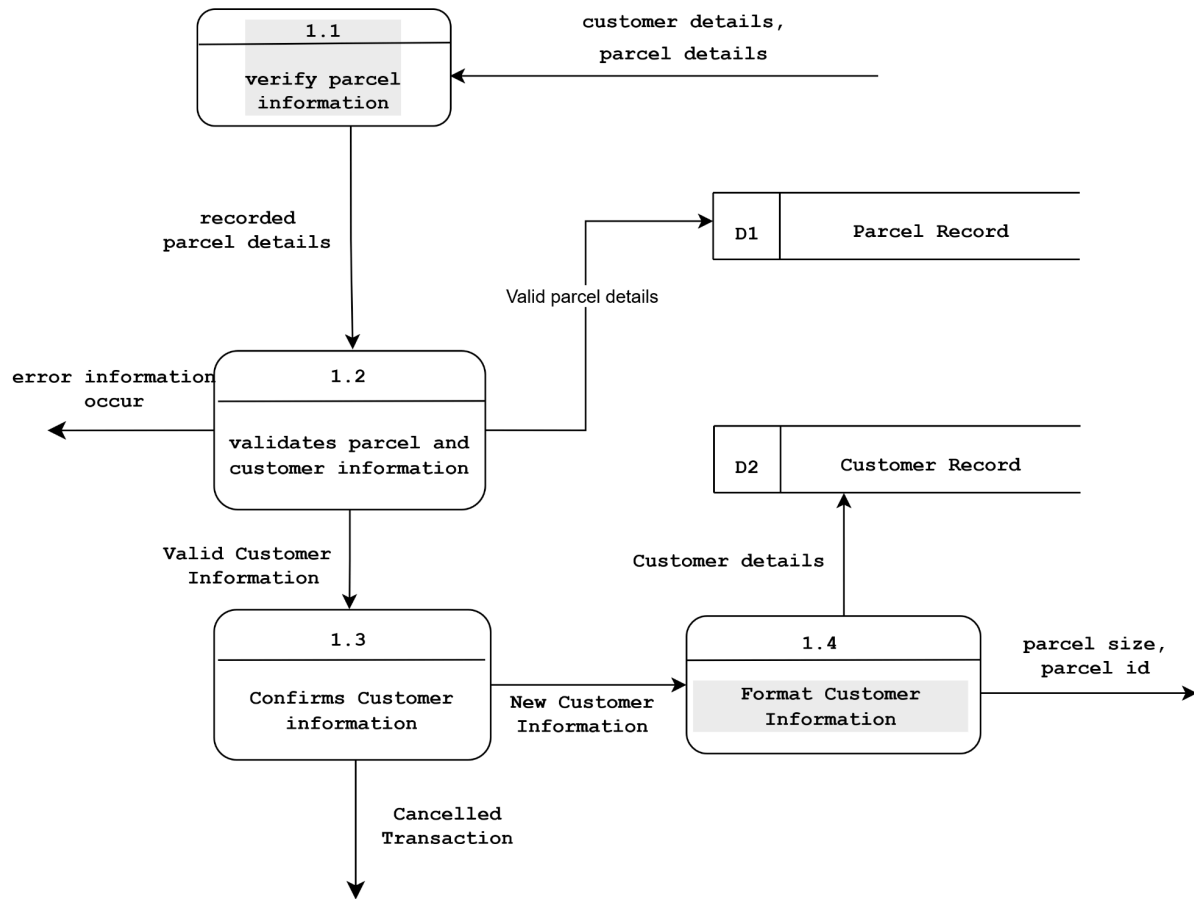


Figure 2.1.3: Child Diagram in process 1 for TO-BE System

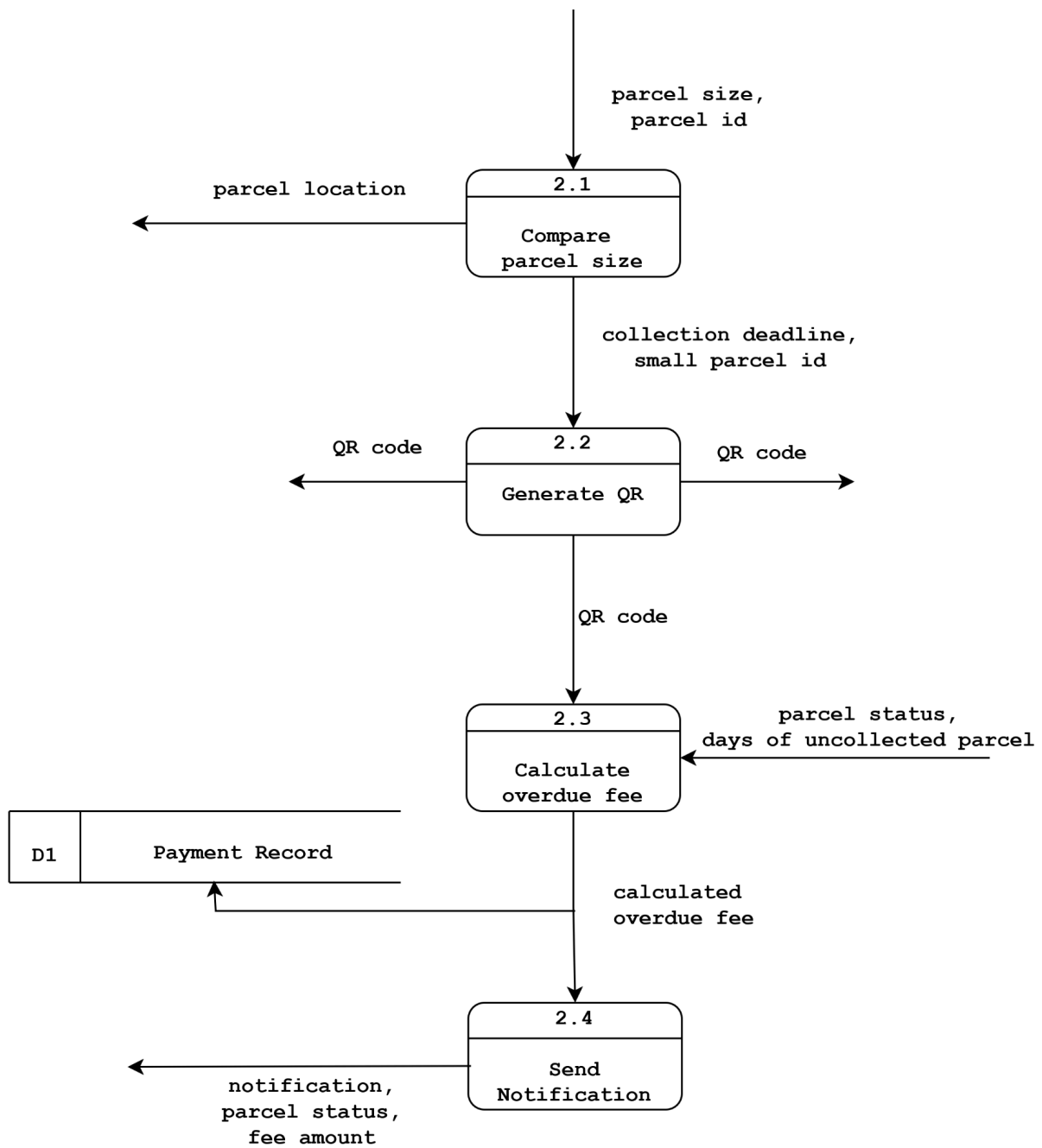


Figure 2.1.4: Child Diagram in process 2 for TO-BE System

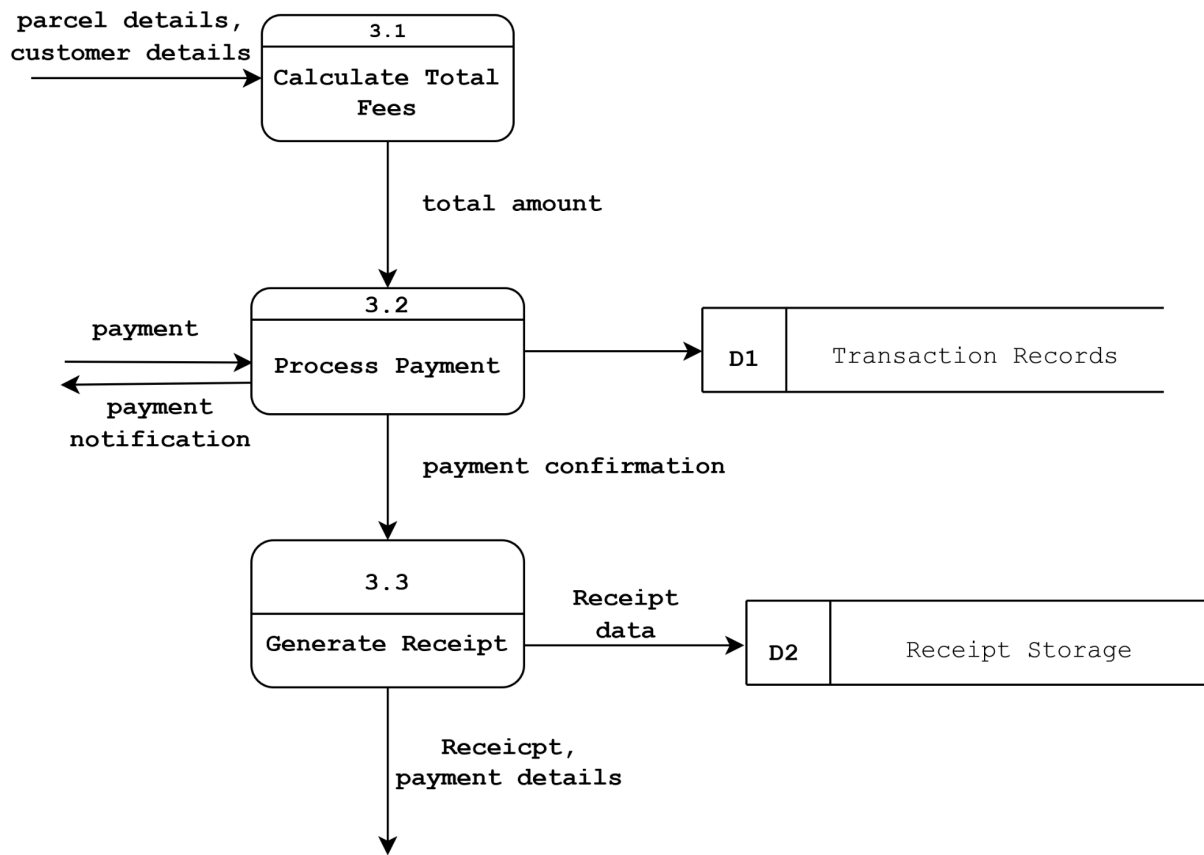


Figure 2.1.5: Child Diagram in process 3 for TO-BE System

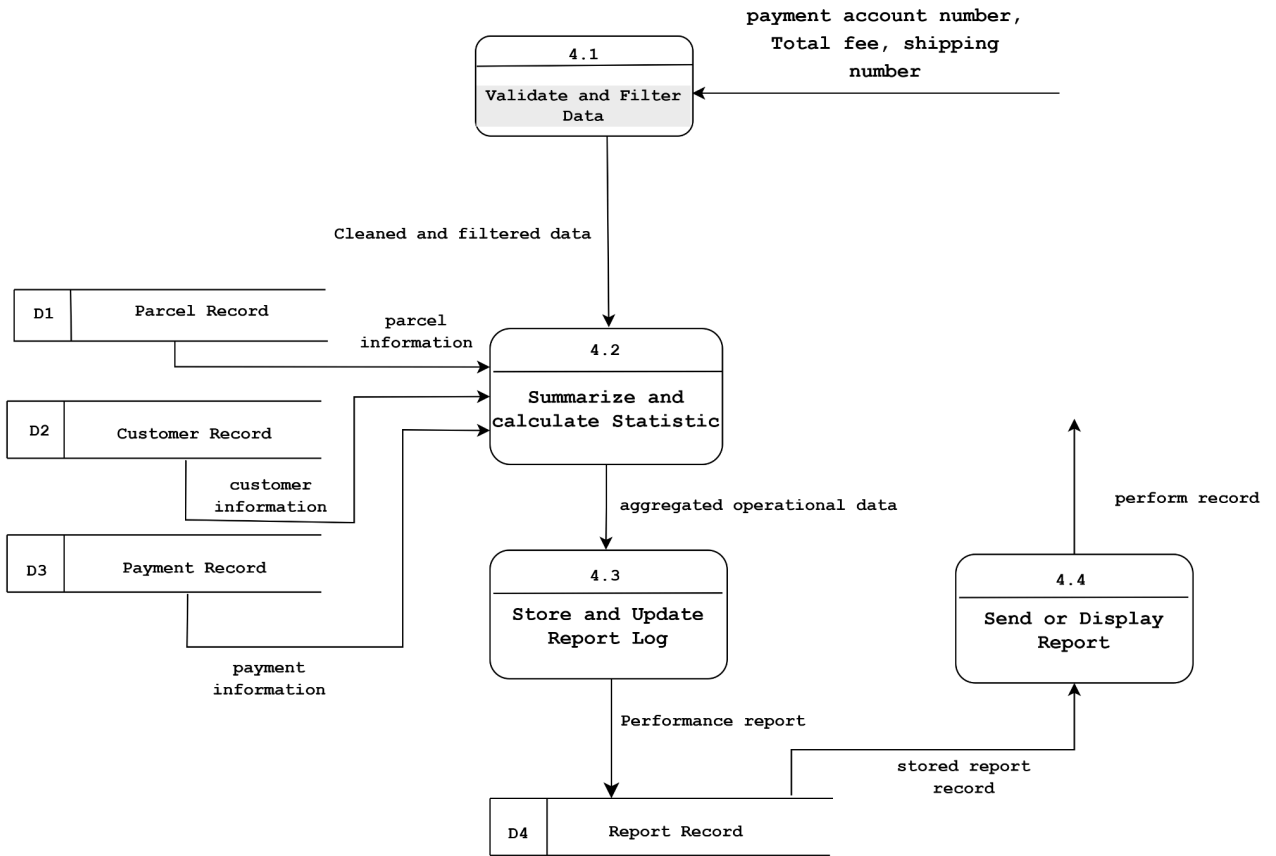


Figure 2.1.6: Child Diagram in process 4 for TO-BE System

2.2 Physical Data Flow Diagram (To-Be)

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Diagram 0 :

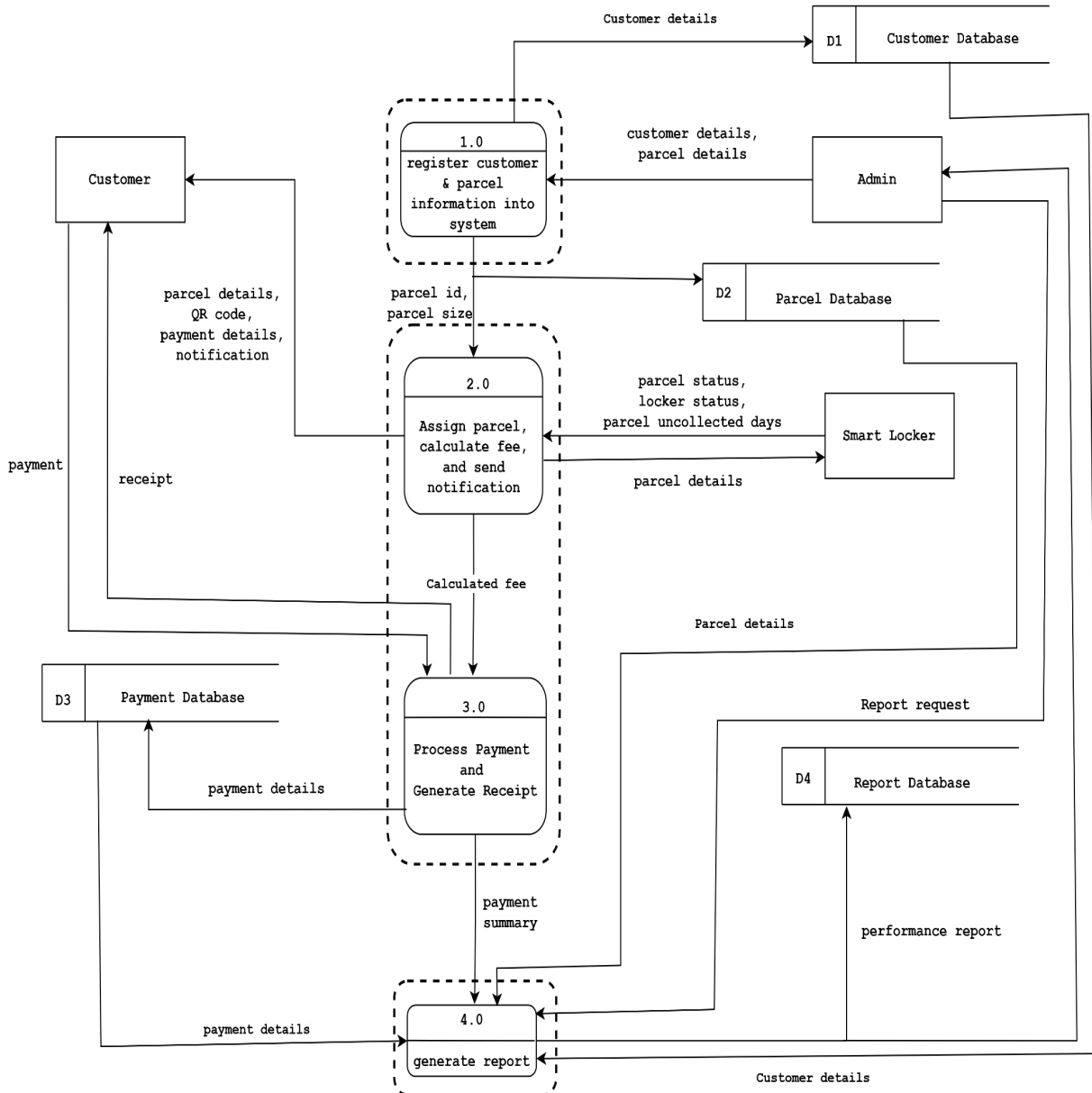


Figure 2.2.1: Diagram 0 for TO-BE System

Child Diagram :

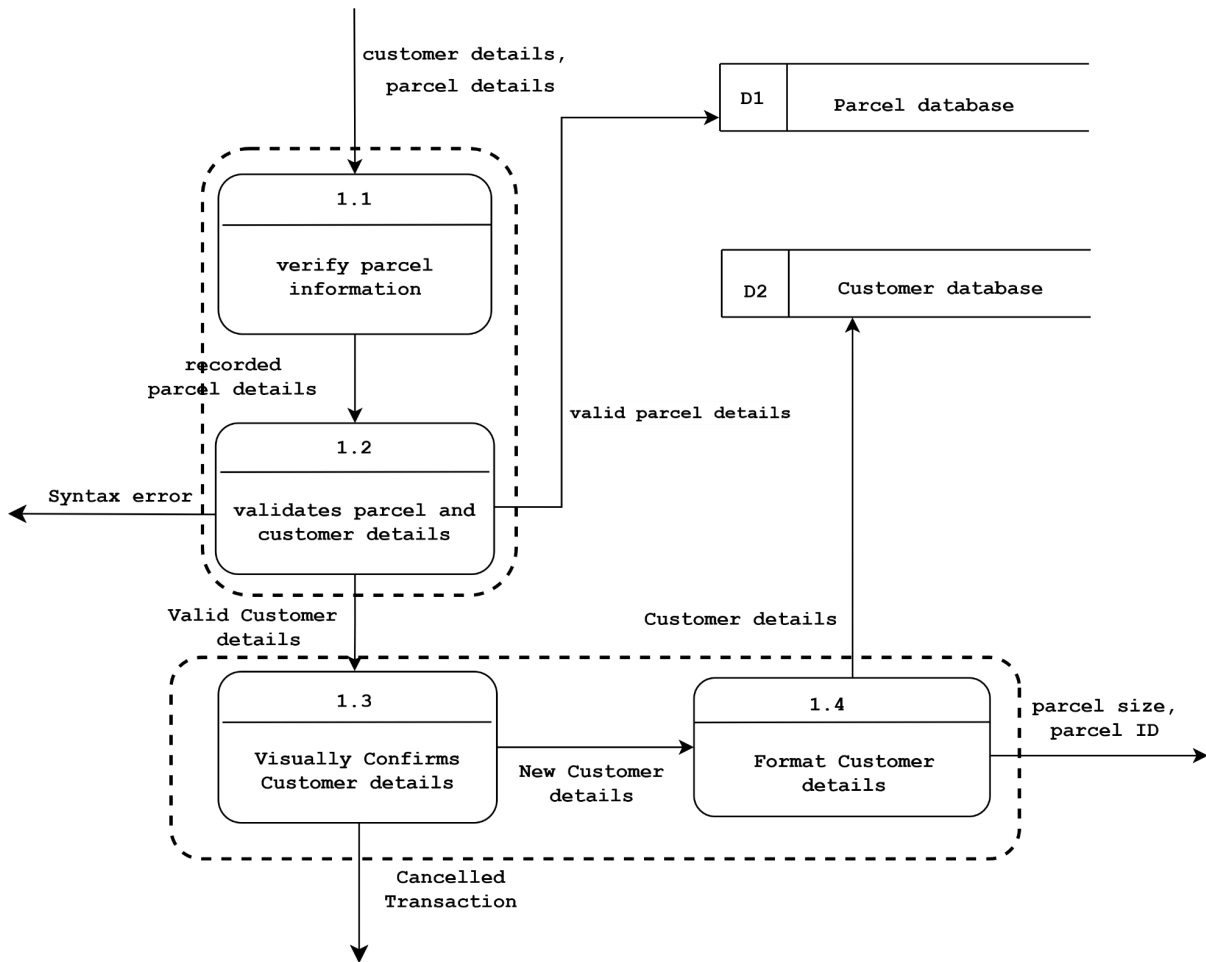


Figure 2.2.2: Child Diagram in process 1 for TO-BE System

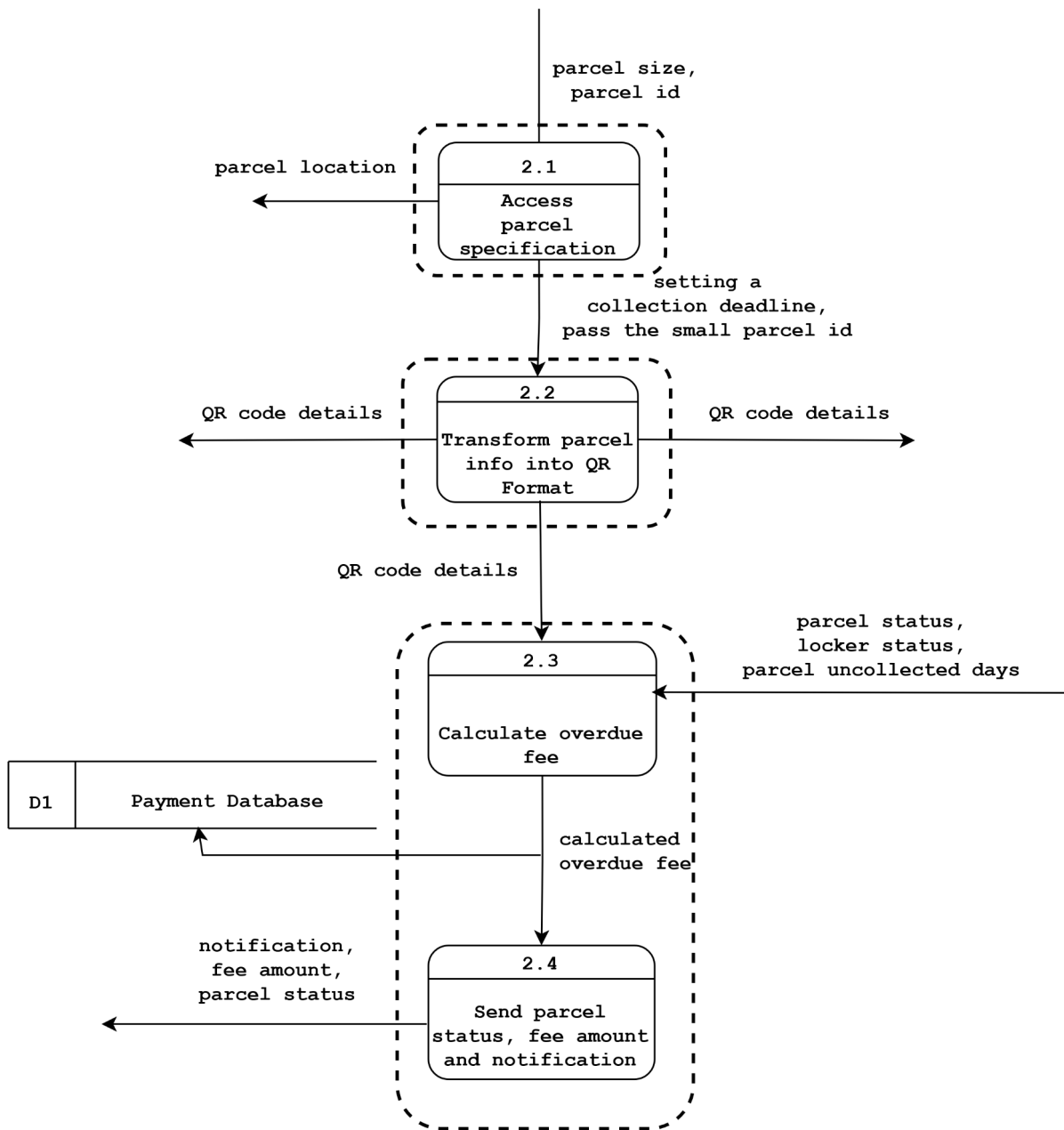


Figure 2.2.3: Child Diagram in process 2 for TO-BE System

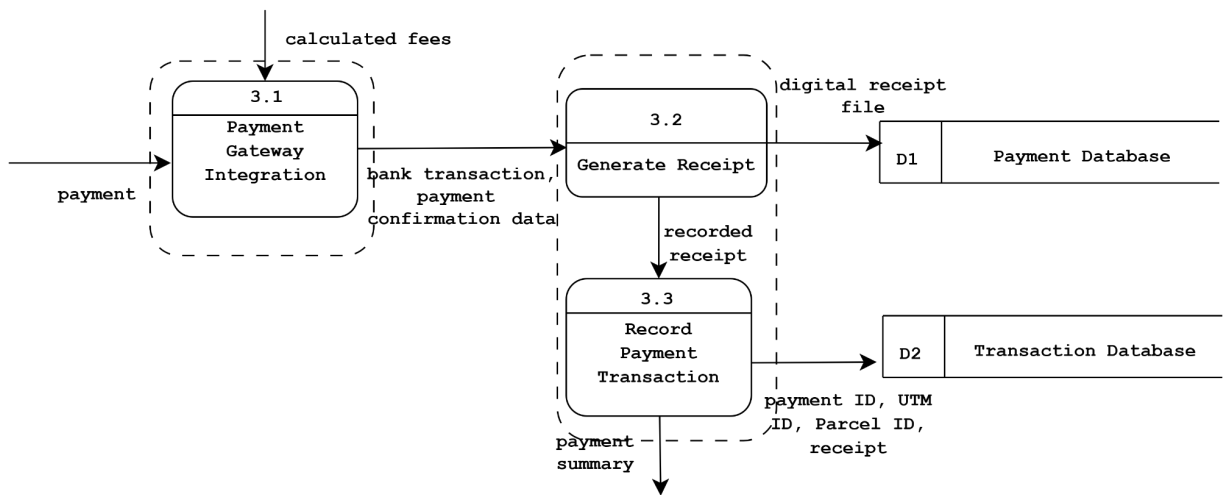


Figure 2.2.4: Child Diagram in process 3 for TO-BE System

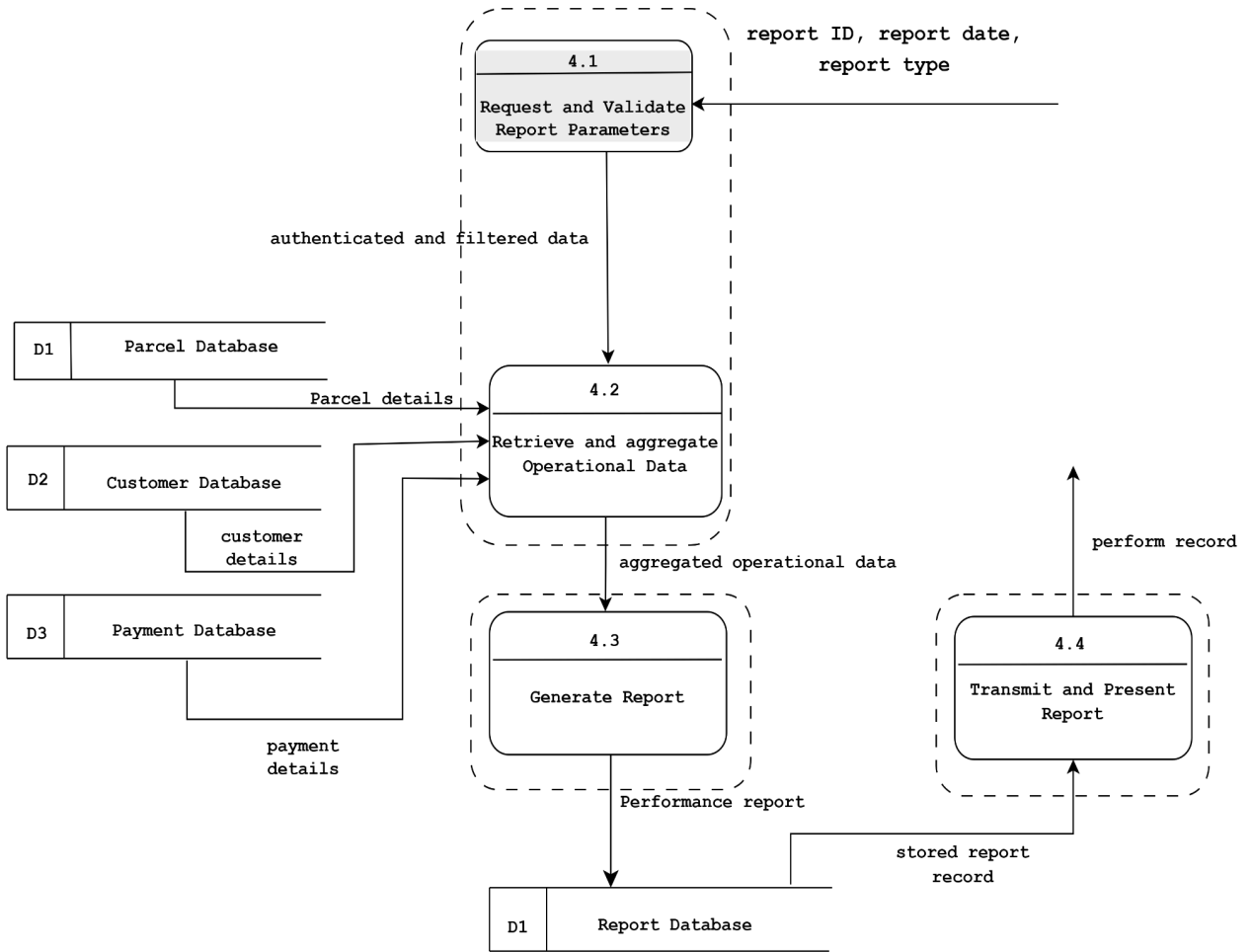


Figure 2.2.5: Child Diagram in process 4 for TO-BE System

3.0 Data & Transaction Requirements

3.1 Proposed Business Rules

3.1.1 Parcel Handling & Logging Rules

- All incoming Parcel must be logged digitally into the central database by an authenticated CPP Staff account.
- A Parcel entry must contain a unique **parcelID**, **trackingNumber**, **courierName**, **parcelSize**, and must be linked to a verified User (recipient) via their **UTMID**.
- The system must automatically assign the Parcel to an available Locker unit based on **ParcelSize** upon logging.
- Each parcel can only be handled by **one staff** and assigned to **one locker** at a time
- If a parcel is collected at the counter, lockerID is NULL.

3.1.2 Customer & Notification Rules

- Once a Parcel is assigned a Locker and its status is '**Ready for Collection**', the system must automatically send an electronic **Notification** to the Customer's registered **phone number**.
- The Customer entity must be specialized into two subtypes, Student and Staff, each inheriting the common **UTMID** but storing subtype-specific attributes.
- Each **Parcel** may have multiple notifications (e.g., Arrival, Ready for Collection, Overdue).

3.1.3 Collection & Transaction Rules

- To collect a parcel, the **Customer** must authenticate via the locker or counter using their **UTMID** and **ParcelID** and scan the QR code linked to parcelID that the system auto-generates.
- A successful Collection using the Locker interface must automatically update the corresponding Parcel record's **parcelStatus** to 'Collected', and the **Locker Status** shall change to "**Available**".
- If a parcel is collected after the allowed period, the system shall automatically record a **penaltyFee** in the **Payment** entity and mark the parcel as "**Overdue**".
- Every payment record must link directly to the corresponding **Parcel**, **Customer**, and the **Staff** who processed it.

3.1.4 Maintenance & Integrity

- Only authenticated Staff accounts can insert, update, or delete Parcel and Payment records. Customers have read-only access to their own parcel history.
- Every creation, update, or deletion of a Parcel record must be linked to the performing Staff's **StaffID** and a precise timestamp for full audit trail.
- If a payment is made, the Payment record must link directly to its Parcel and include the **amount** and **paymentDate**.
- Inactive or completed customer data may be archived only after all related parcels and payments are closed.

3.2 Proposed Data & Transactional

3.2.1 Proposed Data Requirement

Parcel

The data stored in the Parcel entity includes parcel ID, tracking number, courier name, parcel size, arrival date, customer phone number, locker ID, staff ID, and parcel status.

The parcel ID is defined as the primary key as it is unique.

The customer phone number, locker ID, and staff ID are defined as foreign keys referencing the Customer, Smart Locker, and Management relation respectively.

A parcel is received by one customer only. A parcel is logged and managed by only one staff member. A parcel can be stored in one smart locker or handled at the CPP counter. A parcel can make only one payment by a customer. A parcel can appear in one or more system reports whether it is daily, weekly, monthly or yearly.

Smart Locker

The data stored in the Smart Locker entity includes locker ID, customer phone number, parcel status, locker status, capacity, and location.

The locker ID is defined as the primary key.

The locker status used to indicate the availability of the locker, when there is a parcel inside the locker, locker status is “Occupied”, if there’s no parcel inside the locker, locker status is “Available”.

The customer phone number is used as a foreign key, referring to the Customer relation, when a locker is occupied. The parcel status is used as a foreign key from the Parcel relation to indicate whether a parcel in the locker is Arrived, Collected, or Overdue.

A smart locker can store zero or one parcel at a time, but may be assigned to multiple different parcels over time.

Customer (Student / UTM Staff)

The data stored in the Customer entity includes UTM ID, customer name, customer address, and customer phone number.

The UTM ID is defined as the primary key.

The customer name and address are composite attributes. The name consists of first name and last name, while the address consists of college name and block.

A customer can receive zero or many parcels. A customer can receive multiple notifications.

Payment

The data stored in the Payment entity includes payment ID, parcel ID, customer phone number, amount, payment date, parcel status, and staff ID.

The payment ID is the primary key.

The parcel ID, customer phone number, parcel status, and staff ID are foreign keys referencing the Parcel, Customer, and Management relation.

A payment is made for one parcel only, usually due to overdue collection or service charges. A payment is processed by one staff member. A customer can make only one payment for one parcel, forming their payment history. A report can summarize one or many reports at a time.

Management (Admin / CPP Staff)

The data stored in the Management entity includes staff ID, staff IC, staff name, staff role, staff email, staff phone number, and staff login credentials.

The staff ID is the primary key.

A staff member can log parcels, assign lockers, update parcel status, and process payments.

An administrator can generate reports, manage user accounts, and maintain system settings.

A staff member can handle zero or more parcels and payments. An admin/staff can generate zero or more than one report.

Report

The data stored in the Report entity includes report ID, report date, report type, total parcels, pending parcels, collected parcels, overdue parcels, total revenue, and performance summary.

The report ID is the primary key.

Each report is generated by an administrator or staff member.

Reports summarize system performance, including parcel flow, overdue parcel statistics, collection rate, and financial revenue over a specific period. A report can summarize one or many payments and parcels at a time.

Notification

The data stored in the Notification entity includes notification ID, parcel ID, UTMID, notification date, parcel status, and message status.

The notification ID is defined as the primary key as it uniquely identifies each notification record generated by the system.

The parcel ID and UTMID are defined as foreign keys referencing the *Parcel* and *Customer* entities respectively, to ensure referential integrity between parcels and customers.

A parcel can generate one or more notifications. Each notification must be linked to one parcel and one customer only. A **customer** can receive zero or many notifications depending on their parcel activities.

1. Types of Data Currently Stored (To-Be)

- **Customer Information:** UTM ID, customer name, customer address and customer phone number
- **Parcel Details:** parcel ID, tracking number, courier name, parcel size, arrival date, customer phone number, locker ID, staff ID and parcel status (Arrived, Ready to Collect, Collected, Overdue).
- **Notification Record:** notification ID, parcel ID, UTM ID, notification date, parcel status and message status.
- **Payment/Payment History:** payment ID, parcel ID, customer phone number, amount, payment date, parcel status, and staff ID.
- **Staff/Administrator:** staff ID, staff IC, staff name, staff role, staff email, staff phone number, and staff login credentials.
- **Smart Locker:** locker ID, customer phone number, locker status, capacity, and location.
- **Report:** report ID, report date, report type, total parcels, pending parcels, collected parcels, performance summary, overdue parcels, total revenue

2. How Data Is Used in Operations / Reports

Data Type	Usage in Operations / Reports
Customer	Verify student identity, contact for pickup notifications, generate user activity reports.
Parcel	Track parcel progress, assign lockers, update status, generate parcel volume and handle reports.
Smart Locker	Check availability for parcel assignment, update locker status (Occupied or Available), generate locker utilization reports.
Payment	Verify and record penalty payments, generate revenue and outstanding fee reports.
Report	Summarize operational performance, including collection efficiency and financial summaries for management review.
Notification	Automatically inform customers when parcels arrive or are ready for collection, record notification status and time sent, and generate communication efficiency reports.
Management	A staff member can log parcels, assign lockers, update parcel status, and process payments. An administrator can generate reports, manage user accounts, and maintain system settings.

3. User Roles Accessing the Data

User Role	Data Accessed	Purpose
Customer (UTM Staff & Student)	Their own parcel status with locker assignment (QR Code and location) through notification, and payment history.	Check parcel status, receive pickup codes, and view or pay penalties.
CPP Staff	Parcel, Customer, Smart Locker, and Payment data.	Process incoming parcels, assign lockers, update collection status, manage locker maintenance, and record penalty payments.
Administrator	All data (Customer, Parcel, Locker, Payment, Report), summarized Parcel and Payment data.	Full system oversight, generate reports, manage users, troubleshoot issues, and configure system settings , monitor system performance, collection rates, and financial summaries.

4. Problems Solved in Proposed System (Compared to Current System)

Current System(Manual System)	Proposed System Solution
Lack of centralized system: Manual parcel logging in Excel/logbook	Centralized MySQL Database System
WhatsApp notification errors and delays	Automated email/system notification with tracking
No real-time parcel status	Status updated instantly (Arrived → Ready → Collected)
Overdue parcels occupying storage	The system triggers reminders & late penalty payments automatically
Parcel pick-up process not accessibility	Smart locker database for 24/7 auto-collection
No payment tracking	Payment module linked with parcel & staff authentication
No reporting	Auto-generated daily/weekly/monthly reports for admin

3.2.2 Proposed Transactional Requirement

Data Entry

1. Enter the details of a new customer and their parcel such as customer name, UTM ID, phone number, address, shipping number and parcel size to create a new parcel delivery record.
2. Enter the payment details such as collection fee, late fees to create a payment record
3. Enter the finalized payment information such as total fee to generate a receipt record for the customer.
4. Enter the aggregated data to create a performance report record for administrative review.

Data Update/Delete

1. Update the parcel status and customer payment information upon parcel collection, including applying any penalty fees.
2. Update the payment status from 'Pending' to 'Completed' once a payment is successfully processed.
3. Update the smart locker status to 'Available' once a parcel has been collected from it.
4. Delete inactive customer data once collection of parcel has been done and all payments have been settled

Data Queries

1. Display parcel information, collection status, and payment amount
2. Display customer details and parcel collection history
3. Display administrative report information, such as collection rates and payment amount
4. Display smart locker availability status and location for assigning new parcels.

4.0 Database Conceptual Design

4.1 Conceptual ERD

<https://app.diagrams.net/#G10H6xX2TPf60KWcuKL06RD4WMCWtEV7LI#%7B%22pageId%22%3A%226GACXYRyp226Lz3YDWqD%22%7D>

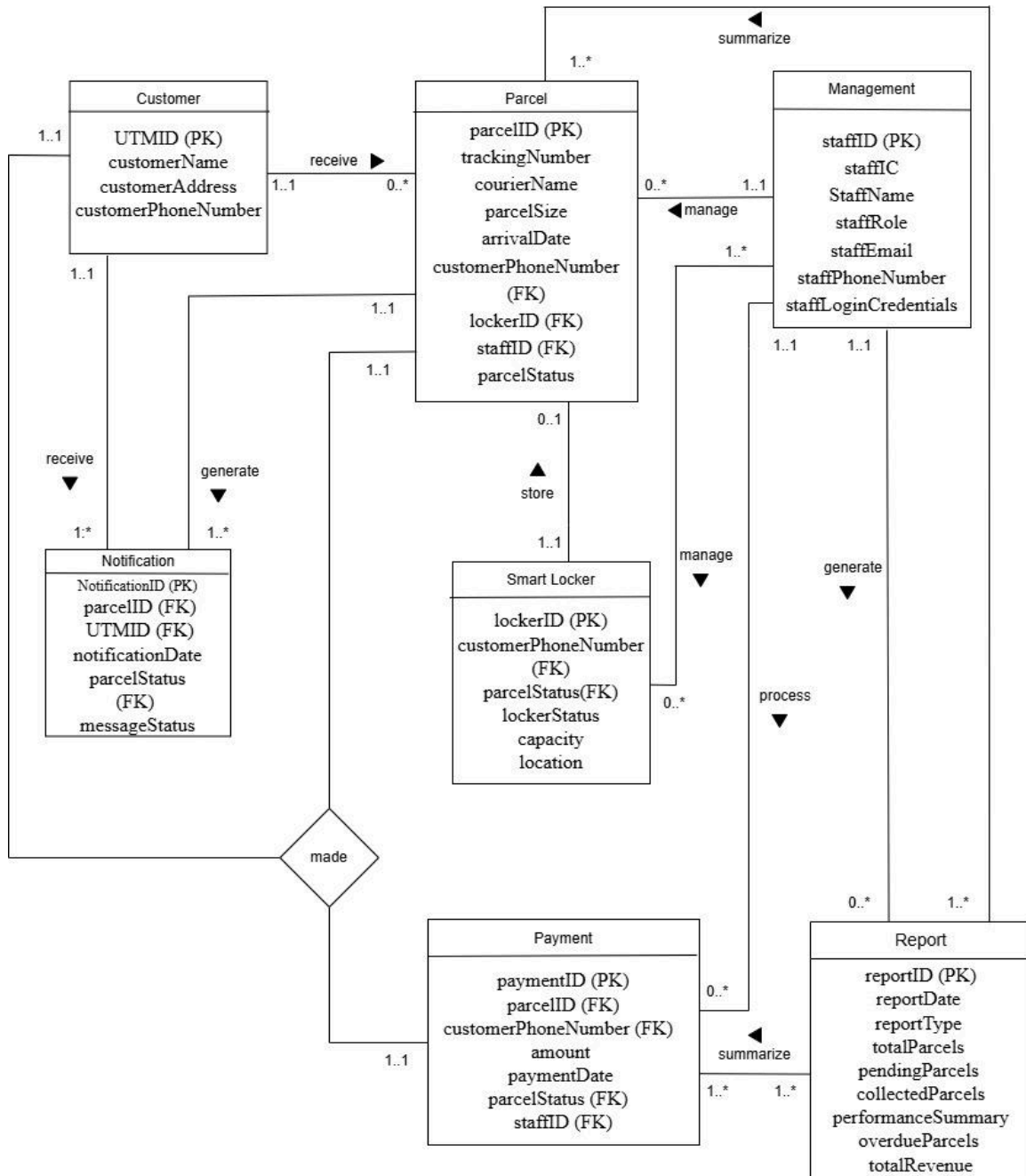


Figure 4.1.1: Conceptual ERD

4.2 Enhanced ERD(EERD)

<https://app.diagrams.net/#G1R6lw7Uw4WbSVEglbng8N-ZJjcHrqUeWH#%7B%22pageId%22%3A%22R2IEEEUBdFMjLhIrx00%22%7D>

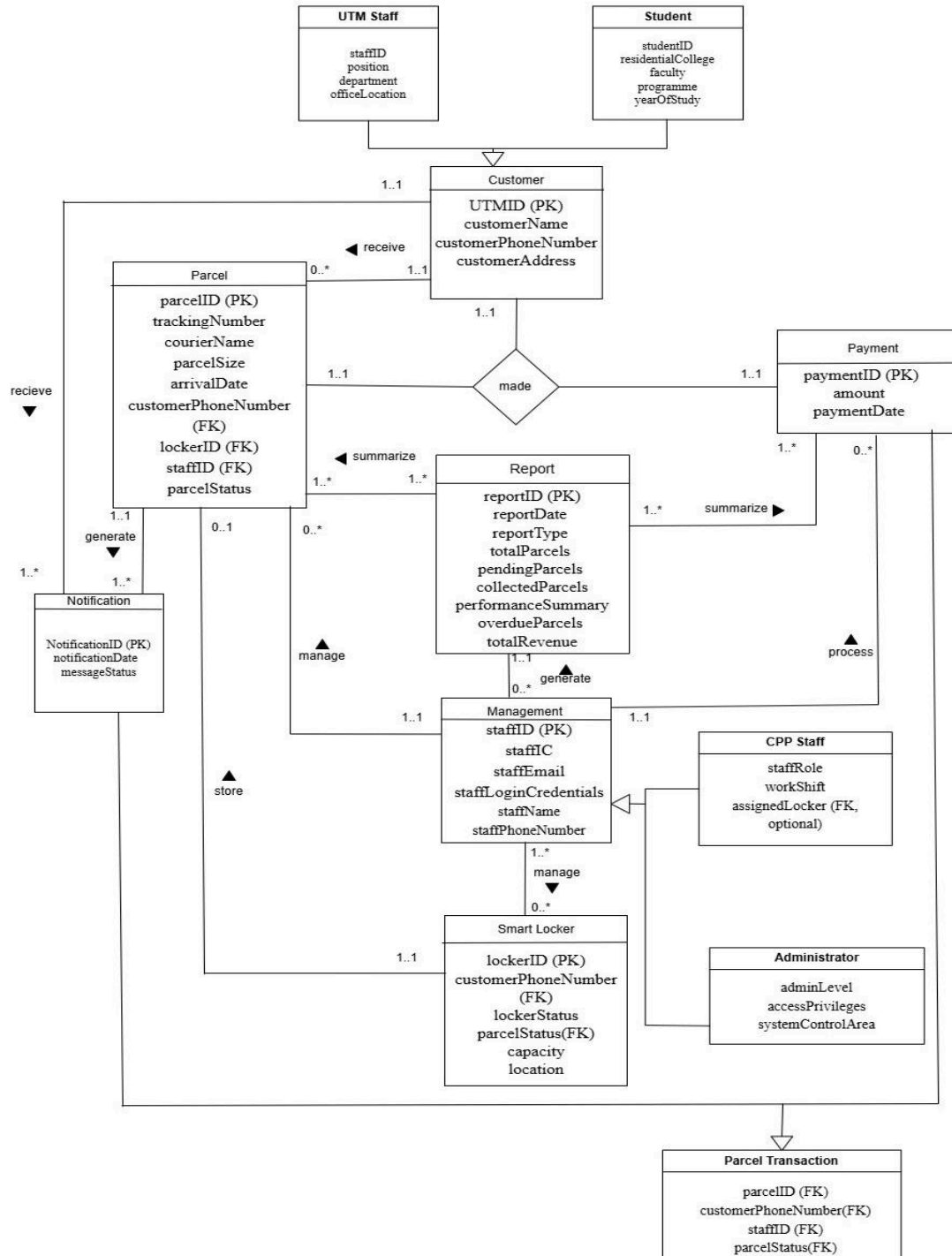


Figure 4.1.1: Enhanced ERD (EERD)

5.0 Data Dictionary

5.1 Description of Entities

Entity	Description	Occurrence
Parcel	Contains information about each parcel handled by CPP, including tracking number, courier name, parcel size, arrival date, and parcel status. Each parcel is linked to one customer, assigned to one smart locker, managed by one CPP staff, and generates a payment for service or overdue.	A parcel belongs to one customer and one locker, and can generate only one payment.
Smart Locker	Records information for all smart lockers used for parcel storage, including locker ID, capacity, parcel status, locker status, and location. Smart lockers are automatically updated when parcels are assigned or collected.	A locker can store zero or one parcel at a time but may be used by many parcels over time.
Customer (Student / UTM Staff)	Stores the details of parcel recipients such as UTM ID, name, address, and contact number. Customers receive parcels and are automatically notified upon parcel arrival or locker assignment. Each customer may receive multiple parcels and notifications.	A customer can receive zero or many parcels, and each parcel belongs to one customer only.
Payment	Stores transaction details for all payments, such as payment ID, amount, payment date, and payment status. Payments are linked to specific parcels and staff members who process them.	A payment is made for one parcel and processed by one staff member. A customer can make only one payment for one parcel.
Management (Admin / CPP Staff)	Contains details of CPP staff and administrators who manage system operations. A staff member can log parcels, assign lockers, update parcel status, and process payments. An administrator can generate reports, manage user accounts, and maintain system settings.	One staff can manage zero or many parcels, payments, smart locker and reports.

Report	Summarizes system performance and operational data such as total parcels, overdue parcels, collected parcels, and total revenue. Generated by the administrator or staff for performance analysis and decision-making.	One report can summarize one or multiple parcel and payment records.
Notification	Records information about notifications sent to customers, including notification ID, UTM ID, parcel ID, date sent, message status, and related parcel status. Used to track and verify that customers are informed about parcel arrivals or ready-for-collection updates.	One parcel can generate multiple notifications, and each notification is linked to one customer.

Entity	Attributes
Parcel	parcelID (PK), trackingNumber, courierName, parcelSize, arrivalDate, customerPhoneNumber (FK), lockerID (FK), staffID (FK) and parcelStatus
Smart Locker	lockerID (PK), customerPhoneNumber (FK), parcelStatus(FK), lockerStatus, capacity, and location
Customer (Student / UTM Staff)	UTMID (PK), customerName, customerAddress and customerPhoneNumber
Payment	paymentID (PK), parcelID (FK), customerPhoneNumber (FK), amount, paymentDate, parcelStatus (FK), and staffID (FK)
Management (Admin / CPP Staff)	staffID (PK), staffIC, staffName, staffRole, staffEmail, staffPhoneNumber, and staffLoginCredentials
Report	reportID (PK), reportDate, reportType, totalParcels, pendingParcels, collectedParcels, performanceSummary, overdueParcels, totalRevenue
Notification	NotificationID (PK), parcelID (FK), UTMID (FK), notificationDate, parcelStatus (FK) and messageStatus

5.2 Description of Relationship

Data Entities	Multiplicity	Relationship	Multiplicity	Data Entities
Customer	1..1	receive	1..*	Notification
	1..1	receive	0..*	Parcel
	1..1	made	1..1	Payment
Management	1..1	manage	0..*	Parcel
	1..*	manage	0..*	Smart Locker
	1..1	generate	0..*	Report
	1..1	process	0..*	Payment
Parcel	1..1	generate	1..*	Notification
	1..1	made	1..1	Payment
Smart Locker	1..1	store	0..1	Parcel
Report	1..*	summarize	1..*	Parcel
	1..*	summarize	1..*	Payment

5.3 Description of Attributes

Entity	Attributes	Description
Parcel	parcelID	A unique, system-generated identifier for the parcel. It is a Primary Key of Parcel entity
	trackingNumber	The unique tracking number assigned by the external courier service
	courierName	The name of the courier company that delivered the parcel.
	parcelSize	The size category of the parcel (e.g., 'Small', 'Large'), used for locker assignment. Small parcel will go to smart locker, big parcel that over locker size will go to CPP counter
	arrivalDate	The date the parcel was officially received and logged into the system by the staff.
	customerPhoneNumber	The phone number of the recipient (Customer) is used for notification. It is a Foreign Key references to Customer(customerPhoneNumber)
	lockerID	The identifier of the specific Smart Locker unit the parcel is assigned to for collection. It is a Foreign Key references to Locker(lockerID)
	staffID	The identifier of the Management member who performed the initial logging of the parcel. It is a Foreign Key references to Management(staffID)
	parcelStatus	The current status of the parcel (e.g., 'Arrived', 'Ready for Collection', 'Collected', 'Overdue')
Smart Locker	lockerID	A unique identifier for each physical Smart Locker unit in the system. It is a Primary Key for Smart Locker entity
	customerPhoneNumber	The phone number of the Customer currently assigned to retrieve the parcel from this locker unit. This field is NULL when the locker is empty. It is a Foreign Key references to Customer(customerPhoneNumber)

	lockerStatus	The status of the locker. If there is parcel inside the locker, status will become “Occupied”, if there is no parcel inside the locker, status will become “Available”
	parcelStatus	It is the Foreign Key references to Parcel(parcelStatus) . The status of the parcel currently inside the locker (e.g., ‘Occupied - Ready to Collect’, ‘Vacant - Collected’).
	capacity	The maximum number of smart locker in UTM
	location	The specific physical location of the smart locker unit on UTM
Customer (Student / UTM Staff)	UTMID	The unique UTM ID of the individual serves as the main identifier for logins. It is a Primary Key for Customer entity
	customerName	The full name of the recipient
	customerAddress	The recipient's on-campus address
	customerPhoneNumber	The primary contact number of the recipient, used for manual lookups and automated notifications
Payment	paymentID	A unique, system-generated identifier for the payment transaction. It is a Primary Key for Payment/Payment History entity
	parcelID	The identifier of the Parcel for which the penalty or fee was incurred, linking the payment directly to the parcel. It is a Foreign Key references to Parcel(ParcelID)
	customerPhoneNumber	The phone number of the Customer who made the payment. It is a Foreign Key for Customer(customerPhoneNumber)
	amount	The total monetary value of the payment transaction
	paymentDate	The date and time the payment was officially recorded
	parcelStatus	The status of the parcel <i>after</i> the payment was made

	staffID	The identifier of the Management member who processed or verified the payment. It is a Foreign Key references to Management(staffID)
Management (Admin / CPP Staff)	staffID (PK)	A unique identifier for the management member. It is a Primary Key for Management entity
	staffIC	The staff member's National Identification Card (IC) number
	staffname	The full name of the staff member
	staffRole	The staff member's role or designation
	staffEmail	The official UTM email address for system communication and verification
	staffPhoneNumber	The staff member's work contact number
	staffLoginCredentials	A hashed and secured representation of the staff member's password or login key
Report	reportID (PK)	Unique identifier for each generated report. It is a Primary Key for Report entity
	reportDate	The date the report was generated or the period it covers
	reportType	The category of the report
	totalParcels	The total number of parcels logged during the reporting period
	pendingParcels	The count of parcels that were available for collection but not yet collected
	collectedParcels	The total number of parcels collected during the reporting period
	performanceSummary	A brief, summarized text field describing the overall performance metrics
	overdueParcels	The count of parcels that have exceeded the allowed collection time
	totalRevenue	The total revenue collected from penalty fees during the reporting period

Notification	NotificationID	A unique, system-generated identifier for the notification instance. It is a Primary Key for Notification entity
	parcelID	The identifier of the Parcel that the notification is about. It is a Foreign Key references to Parcel(parcelID)
	UTMID	The identifier of the Customer (recipient) the notification was sent to. It is a Foreign Key references to Customer(UTMID)
	notificationDate	The notification date that sent to customer
	parcelStatus	The Foreign Key references to Parcel(parcelStatus) . The status of the parcel currently inside the locker (e.g., 'Occupied - Ready to Collect', 'Vacant - Collected').
	messageStatus	The status of the message delivery to customer

Entity	Attribute	Description	Data Type	Null	Multi-Valued
Parcel	parcelID	Uniquely identifies each parcel.(PK)	VARCHAR2(10)	No	No
	trackingNumber	Unique tracking number assigned by courier.	VARCHAR2(20)	No	No
	courierName	Name of the courier company.	VARCHAR2(15)	No	No
	parcelSize	Size of the parcel (Small/Large).	VARCHAR2(10)	No	No
	arrivalDate	Date that parcel was received by CPP.	DATE	No	No
	customerPhoneNumber	Customer's phone number linked to parcel.(FK)	VARCHAR2(15)	No	No
	lockerID	Locker assigned to the parcel.(FK)	VARCHAR2(10)	Yes	No
	staffID	Staff ID who logged the parcel.(FK)	VARCHAR2(10)	No	No
	parcelStatus	Current parcel status (Arrived/Collected /Overdue).	VARCHAR2(20)	No	No
Smart Locker	lockerID	Unique identifier for each locker. (PK)	VARCHAR2(10)	No	No
	customerPhoneNumber	Phone number of customer assigned to locker.(FK)	VARCHAR2(15)	No	No
	parcelStatus	Current status of the parcel in the locker.(FK)	VARCHAR2(20)	Yes	No
	lockerStatus	Locker availability (Available/ Occupied).	VARCHAR2(15)	No	No

	capacity	Maximum number of lockers.	NUMBER(4)	No	No
	location	Physical location of locker.	VARCHAR2(50)	No	Yes
Customer (Student / UTM Staff)	UTMID	Unique UTM identifier for the customer.(PK)	VARCHAR2(10)	No	No
	customerName	Full name of the customer.	VARCHAR2(50)	No	No
	customerAddress	On-campus delivery address (college name and block)	VARCHAR2(50)	No	No
	customerPhoneNumber	Primary contact number of the customer.	VARCHAR2(15)	No	No
Payment	paymentID	Unique payment transaction ID(PK)	VARCHAR2(10)	No	No
	parcelID	Parcel linked to the payment.(FK)	VARCHAR2(10)	No	No
	customerPhoneNumber	Customer making the payment.(FK)	VARCHAR2(15)	No	No
	amount	Total payment amount.	DECIMAL(10,2)	No	No
	paymentDate	Date payment was made.	DATE	No	No
	parcelStatus	Parcel status after payment.(FK)	VARCHAR2(20)	No	No
	staffID	Staff who processed payment.(FK)	VARCHAR2(10)	No	No
Management (Admin / CPP Staff)	staffID	Unique identifier for CPP staff.(PK)	VARCHAR2(10)	No	No
	staffIC	Staff IC number.	VARCHAR2(12)	No	No
	staffName	Full name of staff member.	VARCHAR2(50)	No	No

	staffRole	Role or designation (Admin/CPP Staff)	VARCHAR2(20)	No	No
	staffEmail	Official email for login.	VARCHAR2(50)	No	No
	staffPhoneNumber	Staff contact number.	VARCHAR2(15)	No	Yes
	staffLoginCredentials	Encrypted login credentials.	VARCHAR2(50)	No	No
Report	reportID	Unique report identifier.(PK)	VARCHAR2(10)	No	No
	reportDate	Date report was generated.	DATE	No	No
	reportType	Type of report (Performance/ Revenue).	VARCHAR2(20)	No	Yes
	totalParcels	Total parcels recorded in the report.	NUMBER(5)	No	No
	pendingParcels	Count of pending parcels.	NUMBER(5)	Yes	No
	collectedParcels	Count of collected parcels.	NUMBER(5)	Yes	No
	performanceSummary	Summary of CPP performance.	VARCHAR2(200)	No	No
	overdueParcels	Count of overdue parcels.	NUMBER(5)	Yes	No
	totalRevenue	Total payment revenue collected.	DECIMAL(10,2)	Yes	No
Notification	notificationID	Unique notification record ID.(PK)	VARCHAR2(10)	No	No
	parcelID	Parcel associated with notification.(FK)	VARCHAR2(10)	No	No
	UTMID	Recipient of the notification.(FK)	VARCHAR2(10)	No	No
	notificationDate	Date notification was sent.	DATE	No	No

	parcelStatus	Parcel status in notification.(FK)	VARCHAR2(20)	No	No
	messageStatus	Delivery status of message.	VARCHAR2(20)	No	No

6.0 Summary

In a nutshell, this phase outlines the conceptual database design for the system developed for Cengal Parcel Point (CPP) in Universiti Teknologi Malaysia (UTM). The proposed design emphasizes data organization, accuracy, and relationships between entities and data to support efficient and flexible parcel handling and system automation, reducing human work.

Through the development of the Conceptual ERD and Enhanced ERD (EERD), all essential entities, attributes, and relationships have been identified to represent the operational processes of CPP effectively and clearly. The data dictionary further defines the structure, data types, and constraints, ensuring clarity and consistency in data management.

This conceptual model establishes a solid foundation for the system's implementation phase, where database normalization, integrity, and scalability will be the main goals to be prioritized. Overall, the design contributes to improving service efficiency, reducing manual errors, and enhancing user experience, ultimately supporting CPP's goal of achieving a fully digital and automated parcel management system.